Speaking Through the Prompt: Training-Free Robustness and Expressive Prosody for a Small CPU Persian Text-to-Speech Model

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***Abstract*—Small continuation text-to-speech (TTS) models now run in real time on a laptop CPU, but in practice they fail in two ways that their size and training rule out fixing by retraining: they garble short utterances, and, trained on text stripped of punctuation, they read every question and exclamation as a flat statement. We study both in a public 110M-parameter Persian model and show that the model's second input — the voice prompt it continues — is a control surface that solves both at inference time. *Robustness:* failures depend on how the prompt *ends*. The model was trained to continue the prompt's own utterance, but at inference its text is a new sentence; extending the reference clip with the model's own reading of one neutral sentence makes every prompt end at a finished sentence, so the text reads as the next sentence, which halves failed renders on held-out short sentences (24→12 of 900) and rescues a clip cut mid-clause (44%→3% error) at no per-request cost. *Expressiveness:* the prompt also carries register and pitch range; a re-pitched, widened copy of that reading makes the model itself speak exclamations and surprise livelier (pitch range +38%), while a light rule restores only what no prompt can — the sentence ending — and a loudness plan spans the text. Yes/no questions are then heard as questions three times as often by a recogniser used as a listener, exclamations lose almost no naturalness (UTMOS −0.02 vs. −0.24 for whole-sentence processing), and intelligibility is unchanged. Measurement is part of the contribution: one recogniser overstates short-utterance failures five- to ten-fold, so we judge with two in consensus.**

***Index Terms*—text-to-speech, Persian, voice prompt, prosody, robustness, inference-time methods, low-resource**

I. Introduction

Topic. This paper is about controlling a small, already-trained speech synthesiser without training it. Continuous audio language models generate speech as a sequence of codec latents, conditioned on a short voice prompt that they continue [1]; Pocket TTS distils such a model into a 100M-parameter student that runs faster than real time on a laptop CPU [2], and a community Persian student trained on 973 h of speech is public [3]. We use it, exported to ONNX, as the voice of a Persian–English assistant, and ask what can be fixed at inference time alone (Fig. 1).

Why it matters in theory. A prompted model is told *what* to say by its text and *how* to say it by its prompt, but which properties of the prompt it actually reads is rarely measured. We find that it reads two separate things: where the prompt stops (which sets whether the take begins a new utterance or finishes an old one) and the speaker's register and pitch range (which it copies). A second, more general point concerns training data: a model trained on punctuation-free transcripts learns every prosodic cue that the *words* carry and none that only the *marks* carry, so the missing cues can be located exactly and restored outside the model.

Why it matters in practice. The failures fall where an assistant speaks most — short replies («باشه», «ممنون از پیامتان») and conversational turns that are questions or exclamations. Small CPU models matter where GPUs, cloud TTS and licensed training data are not available; for this model the 24-layer teacher it was distilled from is not even public, so retraining is not an option.

The challenges. (C1) *Short utterances fail*: final syllables are dropped or words garbled, and an engine has no reliable way to notice without a recogniser. (C2) *The model cannot hear punctuation*: «رفتی؟» (“did you go?”) and «رفتی.» (“you went.”) reach it as the same tokens, so every yes/no question is read as a statement, exclamations sound like news, and loudness never changes from sentence to sentence. Behind both lies a measurement problem: the usual recogniser judge hallucinates on short audio, and there is no cheap judge of prosody.

What we propose. (i) *Prompt self-extension:* once per voice the model reads one neutral sentence; every later take is conditioned on the reference clip followed by that reading, so the prompt ends at a finished sentence and the text it is given can be its next sentence (C1). (ii) *Prompt styling:* the same reading, re-pitched, widened and made louder by signal processing, is the prompt for exclamations and surprised questions, so the model itself speaks them higher and livelier; only the prompt is processed (C2). (iii) *An ending rule and a loudness plan* that restore, from the written mark, only what no prompt can carry: the final rise of a yes/no question, the suspended end of wh-questions and exclamations, sustained energy and sentence-to-sentence loudness (C2). (iv) *Measurement:* a two-recogniser

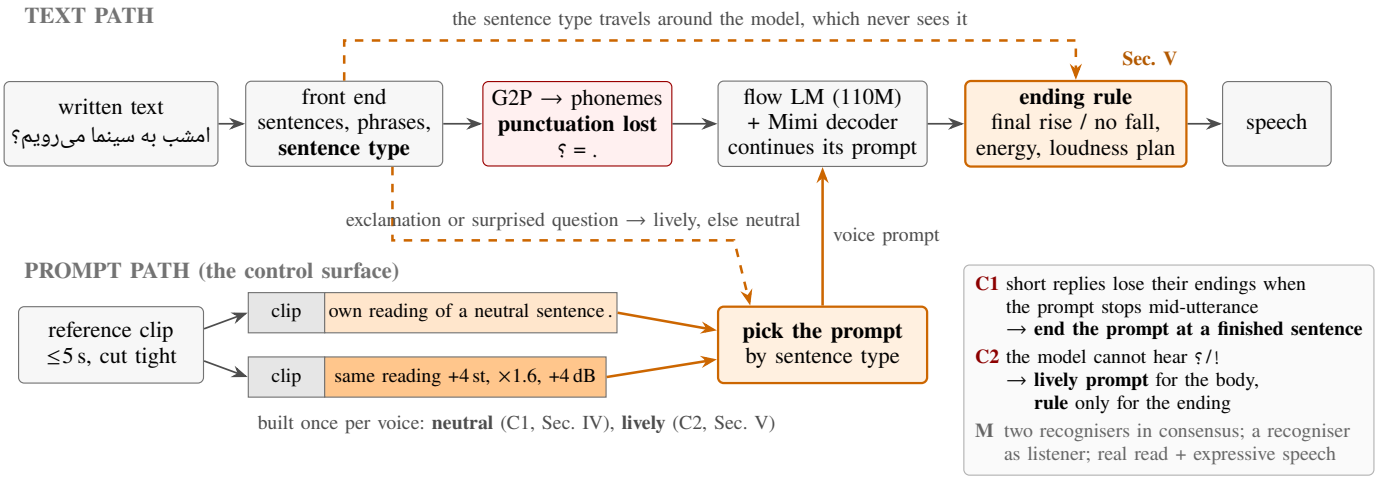


Fig. 1. Speaking through the prompt. The model has two inputs: phonemes, from which punctuation is lost, and a voice prompt that it continues. We use the prompt as the control surface. Ending it at the model's own finished sentence fixes short utterances (C1, Sec. IV); a livelier copy of that sentence, chosen per sentence type, gives exclamations and surprise their delivery, and a rule restores only the sentence ending the model cannot know (C2, Sec. V). The sentence type travels around the model in the text layer. Orange: our components.

consensus judge, a recogniser used as a listener for sentence type, and real read and expressive Persian speech as references.

Results against existing work. Prior fixes for robustness train the model — preference alignment [4], short-phrase fine-tuning [5], distilled ASR-verified best-of-N [6] — or verify each take with a recogniser; ours needs neither and halves failed short renders while dropping retries, which halves generation work. Prior expressive control chooses expressive recordings as prompts [7] or trains a separate style prompt [8]; we make the expressive prompt from the model's own speech and add only boundary tones, which triples the rate at which yes/no questions are heard as questions over a contour rule alone and removes most of its quality cost. All code, sentence sets and evaluation scripts are public: https://github.com/abolfazltorbat/persian_tts.

II. Related Work

Robust autoregressive TTS. Hallucinated, repeated and dropped content is a known failure of autoregressive TTS [4], [9]; remedies include ASR self-verification and distillation [6], fine-tuning on 1–2-word inputs [5] and repetition penalties [10], [11]. For long-form speech, attention steering [12] and boundary-aware post-training [13] help, and the acoustic prompt must stay in context [14]. We vary the prompt itself: how it ends.

Sentence type and expressiveness. In English TTS a final question mark is the main lever on the nuclear tune, which can be steered through hidden states [15]; TTS systems underproduce rises on yes/no questions and over-produce them on wh-questions [16]; long prompts help reproduce a speaker's intonation [17]. Expressive Prompting selects intense prompts from a pool of recordings [7], and IndexTTS2 separates style from timbre prompts [8]. In Persian, yes/no questions end in a high boundary tone, while wh-questions carry the nuclear accent on the question word and fall [18]. Pitch is modified by TD-PSOLA [19] on YIN pitch tracks [20].

Persian speech resources. ManaTTS [21], ParsVoice [22], homograph-aware G2P [23], the ShEMO emotional speech corpus [24] and PARSA-Bench [25].

III. Model, Engine and Measurement

Model and engine. pocket-tts-farsi-v2 [3] is a 6-layer student (110M parameters) over 24 kHz Mimi latents at 12.5 Hz [26]. Each generation unit is `[BOS, voice prompt, phonemes, latents…]`. In training, each utterance was cut at a random boundary between two aligned words: the audio before the cut (at most 5 s) is the prompt, and the target is the *rest of the same utterance* with its remaining words as text [3]. The model thus learns to continue the prompt's utterance; at inference the text is a new sentence instead. A front end splits text into sentences and phrases, keeps each sentence's written type, and packs phonemes into units of at most 30 tokens; the G2P drops punctuation, and the training transcripts were phonemised with «؟» removed (it shares a symbol with the glottal stop), so the model has no sentence mark at all.

Two recognisers in consensus. Whisper large-v3-turbo [27] is a poor judge of short audio: on 76 *real* short Persian recordings (1.8 s) it fails 17% of correct clips, often writing «موسیقی» ("music") for speech, while a Persian wav2vec2-CTC recogniser fails 10.5%; on synthetic short takes the two correlate only 0.23. A take therefore counts as failed (consonant-skeleton CER > 0.3, short vowels being unwritten in Persian) only if *both* fail it. Under this label 1.3% of 2,070 Persian development takes fail, all in units of at most six phoneme tokens; Whisper alone reports 12.3% overall and 41% for 1–3-token units.

A recogniser as listener. Whisper writes punctuation. For a yes/no question without a question particle the words give

it no clue, so a final «؟» in its transcript means it *heard* a question. On such questions it writes «؟» for 39% (12/31) of expressive real speech (ShEMO radio plays [24]), 5% (1/19) of read speech (GPTInformal, one speaker) and 3% of real statements — a coarse but reference-free judge of sentence type.

Other measures. Naturalness by UTMOS [28] and DNS-MOS SIG [29], used only relatively; speaker similarity to the clip by WeSpeaker [30]; F0 by pYIN [31] in semitones (st) re each utterance's median; energy in dB. Paired differences carry 95% case-bootstrap confidence intervals. Development and test sentences are disjoint, and every choice was made on development data.

IV. Robustness: How the Prompt Ends

Post-processing cannot see failed takes. Before changing the input we asked whether an engine can reject bad takes without a recogniser. Signals it has for free — level, speech fraction, duration per vowel, end-of-speech timing, syllable nuclei vs. vowels, self- and prompt-similarity of the latents — predict which *units* are risky (case-grouped cross-validated AUC 0.88, mostly unit length) but not which *take* of a unit failed (best within-unit AUC 0.67). They are not blind in general: takes that replay the prompt are separated with AUC 0.98. Natural failures are short deletions, and these they miss. The engine's own retry rule rejects 29% of takes, 96% of them good, and buys 2.35%→2.15% mean error for 49% more generation. The fix has to come before generation.

Prompt self-extension. In training the text always continues the prompt's utterance; at inference it never does. A reference clip cut right after its last phoneme, usually mid-utterance, therefore asks for a continuation of *its* utterance that the given words cannot provide, and short targets, with the least room to recover, suffer most. Among training cuts, only those that happen to fall at a sentence boundary have a new sentence as their continuation; a prompt that ends at a finished sentence puts every inference into that case. We do not claim that training prompts typically end at sentences — most random word boundaries fall inside one — but that this is the subset whose continuation matches ours. Once per voice, the model reads one neutral sentence («امروز هوا کمی سرد است و باد می‌وزد.», "it is a little cold today and the wind blows") from the plain clip four times with fixed seeds; the take of median speech length (no recogniser) is kept, as raw latents, up to its speech end plus one frame. Every later unit is conditioned on the clip followed by the last (5 s − |clip|) of this extension. The prompt is encoded once into the key–value cache that every unit already copies, so a request costs nothing extra.

It is the ending that matters (Fig. 2a). On 30 short development sentences × 6 seeds for each of two voices (male 1.36 s and female 2.0 s clip; 360 paired takes per layout), the extension cuts failures from 25 to 10. Repeating the clip, which adds speech but keeps the abrupt ending, does not help (35); the same extension cut after two thirds — same voice, same latents, unfinished — multiplies failures (120); and with the clip placed last the model reads the extension aloud instead of the text in 94–99% of takes («بله» comes out as «امروز هوا کمی سرد ...»). Teacher-forcing a carrier word, or priming each unit with the previous unit's tail, adds nothing. Which sentence is read matters (two of seven candidates helped both voices) and was fixed on development data.

TABLE I
Prompt self-extension, held-out (paired renders; change with 95% CI)

measure	clip	clip + ext.	change [CI]
short set: 50 fa + 10 Eng./mixed sentences × 6 seeds × 3 voices			
failed renders (of 900 fa)	24	12	
consensus CER	.032	.022	−.011 [−.017, −.004]
default voice (1.36 s clip)	.060	.039	−.021 [−.036, −.008]
UTMOS / spk. change (1,080)			+.13 / −.012
speaker-matched extension take			+.18 / +.009
3 new real clips cut 0.02 s after speech; 50 fa sentences, 3 seeds			
failed renders (of 450)	96	19	−.167 [−.233, −.106]
clip ending mid-clause, CER	.441	.030	−.411 [−.546, −.288]
bench: 79 mixed cases × 3 seeds (55 fa cases scored)			
consensus CER	.029	.023	−.006 [−.010, −.001]
pitch reset at chunk joins (st)	5.25	3.92	−1.33 [−1.95, −.77]
default voice, one take instead of the 3-take retry loop			
consensus CER	.060	.041	−.019 [−.033, −.006]
generations per unit	2.1	1.0	

Held-out results (Table I, Fig. 2b). On 60 unseen short sentences rendered through the full production path, failed renders halve and UTMOS rises for every voice. The prediction of the mechanism holds on three *new* real clips cut 0.02 s after their last sound: the sentence-final clip improves slightly, the comma clip halves its error, and the mid-clause clip, nearly unusable alone, becomes ordinary — although the mid-clause clip ends at a word boundary just as training prompts do, which is why we place the mechanism in what the text must continue, not in the cut itself. On a mixed bench of long and code-mixed text, error also falls, and units of one sentence sound less like separate sentences (smaller pitch reset at joins). With the extension a single take beats the engine's three-take loop, so retries can be dropped and the generation work halves; on 40 assistant-style replies this configuration is no worse on any measure with 28% fewer generations. The cost is speaker similarity on one voice (female, −0.03); choosing the extension take by speaker similarity removes it (+0.009 pooled) at part of the intelligibility gain. English sentences, read through mapped Persian phonemes, keep the plain clip.

A first listening check. A native Persian listener compared plain and extended prompts blind: 60 pairs of held-out short sentences in the default voice, same seed, left/right at random, choosing the better reading or "same" and marking any word error. Preference did not differ (same 33, plain 15, extended 12; sign test $p = .70$); word errors were marked in 7 plain and 5 extended renders (3 vs. 1 on Persian-only sentences). None of the 12 takes the listener marked was among the 9 the recognisers failed: errors on such short sentences are rare and idiosyncratic, so the recogniser-measured gain needs a larger listening test to be confirmed by ear.

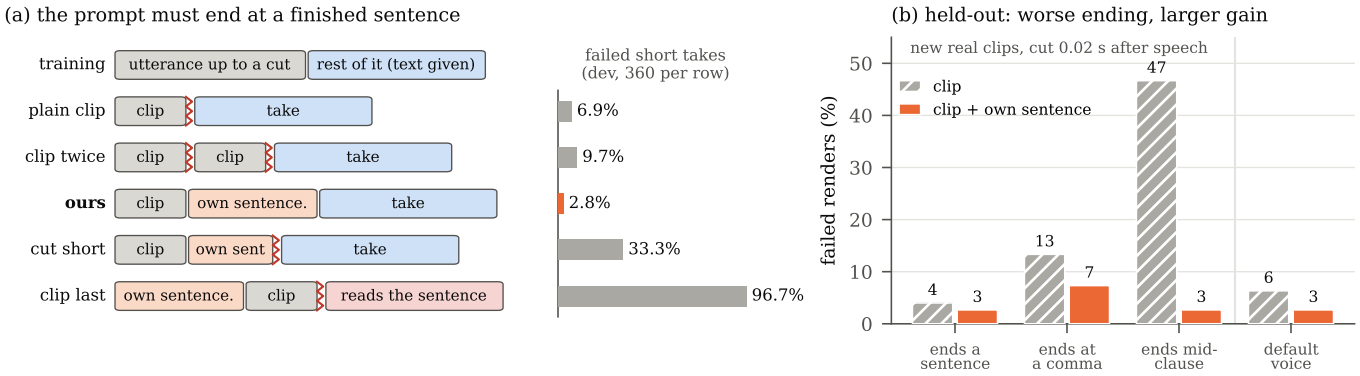


Fig. 2. How the prompt ends decides whether a short sentence survives. (a) Prompt layouts (zig-zag: the prompt stops mid-utterance) and the share of failed short takes on development units (30 sentences × 6 seeds × 2 voices = 360 per row, consensus judge). In training the text continues the prompt's utterance. Only a prompt ending at a finished sentence helps; more of the same clip does not, the same extension cut short multiplies failures, and with the clip last the model reads the extension aloud. (b) Held-out: new real clips cut 0.02 s after their last sound, by where they end, and the engine's default voice; the less finished the clip's ending, the larger the gain.

V. Expressiveness: What the Prompt Carries

Diagnosis. Because the model never sees a mark, a yes/no question and its statement twin are the same audio sample for sample (final-pitch difference 0.00 st; 4.75 st in real speech). What the words carry, the model does reproduce: on wh-questions it accents the question word (+3.4 st; real +3.8). Real speech also shows what else is missing (Fig. 3e, f): renders of exclamations lose 5.7 dB of energy from their first to their second half, while ShEMO's neutral, surprised and happy speech loses 4.0, 1.8 and 0.6 dB; real wh-questions end level (−0.9 to −1.4 st) but ours fall like statements (−4.4); and Persian alternative questions («چای یا قهوه؟») fall. A first remedy — reshaping the whole contour of questions and exclamations by TD-PSOLA — gives questions a final rise but costs 0.24 UTMOS on exclamations, an audible roughness, and is barely heard: Whisper writes «؟» for 4% of such questions. Looking for a pitch direction inside the model failed: a linear F0 probe on Mimi latents explains 30% of the variance, but pushing latents along it changed nothing until the sound broke.

The prompt carries register and pitch range. Since the model continues its prompt, we let the prompt do the expressive part. The self-extension is decoded, its F0 raised by r st with excursions scaled by k (TD-PSOLA), its level raised by g dB, and it is encoded back with the Mimi encoder; the take is generated from the clip followed by this *styled* extension and decoded as usual. On 108 development renders the take follows the prompt's register — from a quarter to all of the shift, least for the female voice (Fig. 3d) — and with $(r, k, g) = (4, 1.6, 4)$ its pitch range widens from 9.0 to 12.2 st with consensus error, UTMOS, DNSMOS SIG and speaker similarity unchanged (−0.001, +0.07, +0.01, −0.001). A decode–encode round trip alone moves register by at most 0.5 st. Stronger styling begins to cost intelligibility, and for a voice whose extension lies above 165 Hz the register shift is halved (on the female voice $r = 4$ cost 0.40 UTMOS on exclamations, $r = 2$ 0.12). This *lively* prompt is used for exclamations and surprised questions; a milder one for all questions widened the range so much that it deepened the fall before the final rise (yes/no heard as questions 12%→7%) and was dropped. Each styled prompt is built once per voice (0.5 s).

The rule restores only the ending. From its written mark, interjections and question words the front end types each sentence: statement, yes/no question (with tag questions), wh-question, alternative question, surprised or echo question («جدی؟», «!؟»), and exclamation with an intensity that grows with «!!» and interjections. A yes/no question rises over its *last syllable* only — found as the energy valley before the last vowel nucleus — along $u^{1.3}$ to 9 st above the median, and that syllable is lengthened by 15%; a surprised question rises to 8 st; the final fall of wh-questions and exclamations is stopped at −1 st; alternative questions are left to fall; and the second half of questions and exclamations is raised by up to 2 and 3.5 dB with a plain gain envelope, as in excited speech. Over a text, exclamations gain +3 dB times their intensity, surprised questions +2, other questions +0.5, and a paragraph's first and last sentences +1 and −1.5 dB; a look-ahead peak limiter keeps the plan intact where scaling a whole sentence down would undo it. These settings were chosen on development sentences by the recogniser-heard question rate against UTMOS.

Results (Table II, Fig. 3). Yes/no questions are heard as questions three times as often as with the contour rule (12% vs. 4%; read speech 5%, expressive 39%) at the same rise over their statement twins; alternative questions keep their fall; wh-questions end 1.4 st higher, toward real speech; surprised questions are spoken 3.7 st higher and 1.9 dB louder; exclamations come out with a 38% wider pitch range, 2.2 dB louder, with their energy kept up between real neutral and excited speech, at almost no UTMOS cost (−0.02 against −0.24). Through whole texts, sentence loudness varies twice as much as before (0.67→1.31 dB; read speech 1.52) and register a third more. Intelligibility is unchanged. The costs are a small loss of speaker similarity on exclamations (−0.028) and of UTMOS

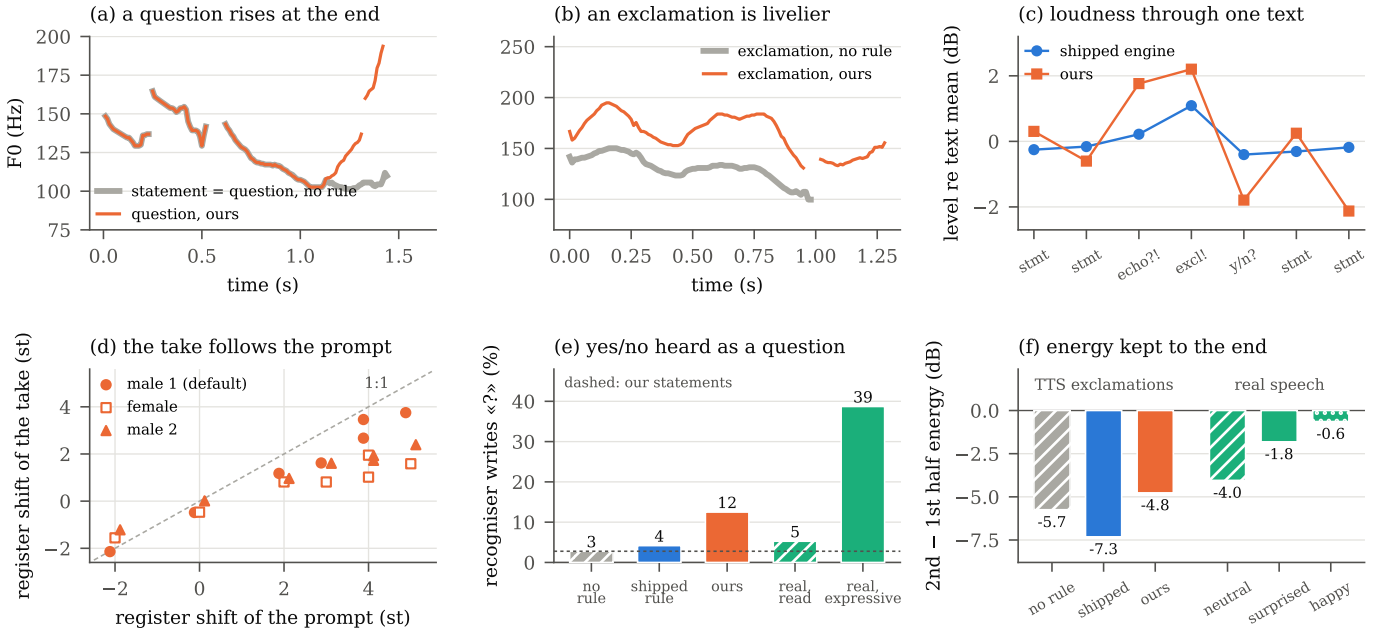


Fig. 3. Expressiveness from the prompt, endings from the rule (held-out sentences unless noted). (a) A yes/no question and its statement twin are one rendering for the model; the rule adds the final rise. (b) An exclamation generated from the lively prompt: higher and wider throughout, with no processing of its body. (c) Sentence loudness through one text, shipped engine vs. ours. (d) Register of the take against the register shift of the styled prompt (development, three voices). (e) Yes/no questions heard as questions by Whisper, vs. real read and expressive speech. (f) Energy of the second half of a sentence relative to the first: our exclamations against real speech by emotion (ShEMO).

TABLE II
EXPRESSIVENESS, HELD-OUT (3 VOICES × 2 SEEDS; 348 QUESTION AND EXCLAMATION RENDERS) AND WHOLE TEXTS (8 TEXTS, 48 RENDERS). SHIPPED: CONTOUR RULE ALONE; OURS: LIVELY PROMPT + ENDING RULE + PLAN

measure	no rule	shipped	ours
yes/no heard as question («؟»)	3%	4%	12%
yes/no − statement, final F0 (st)	0.00	+3.63	+3.56
alt. questions, final F0 (st)	−0.40	+2.68	−0.40
wh-questions, final F0 (st)	−4.35	−4.35	−2.99
surprised q., register (st)	0	+1.05	+3.65
exclam., pitch range (st)	7.5	7.1	10.4
exclam., level (dB)	0	+0.7	+2.2
exclam., 2nd−1st half energy (dB)	−5.7	−7.3	−4.8
texts: sentence level SD (dB)	0.49	0.67	1.31
texts: sentence register SD (st)	1.22	1.20	1.61
consensus CER (fails)	.041 (9)	.043 (10)	.041 (8)
ΔUTMOS, all / exclam.		−.12 / −.24	−.10 / −.02
ΔSIG / Δspeaker sim.		−.028 / −.002	−.040 / −.009
texts, ours−shipped: CER/UTMOS			−.001 / −.03

on surprised one- and two-word questions (−0.25), where both the rise and the prompt act on very little audio.

What did not improve clarity. The default voice renders 10 dB weaker at 2.5–6 kHz than its own clip, but equalising each render toward the clip lowered UTMOS and DNSMOS SIG (−0.06 each); equalising the prompt instead brightened the take by 4.5 dB at a cost in speaker similarity; pitch marks from zero-frequency filtering [32] made TD-PSOLA rougher on codec speech (UTMOS −0.37 vs. −0.27 for a flat +2 st shift); and re-encoding processed audio with the codec hid the processing cost but cost 2.1 points of error. The clarity gain that held up was structural: moving the expressive work from the signal into the prompt.

Listening test. To check these acoustic and recogniser-based findings by ear we built a blind, 15-minute listening test for native Persian listeners (page and scoring script in the repository). *Part 1* plays single clips with the text shown *without* its final mark; the listener labels each a statement, a question or an exclamation/surprise and rates its naturalness (MOS, 1–5). Minimal pairs are split so that a listener hears only one member: the statement, or the question in one system's version. *Part 2* compares the shipped and our versions of 8 whole texts and 8 sentences (CMOS, −3..+3, for naturalness; which one conveys the punctuation better). *Part 3* rates 24 short replies, plain vs. extended prompt, for clarity (1–5) and word errors. Assignments follow a Latin square over listeners; we report bootstrap CIs over listeners, Wilson intervals for identification rates, sign tests, and inter-rater agreement (Krippendorff's α for ratings, Fleiss' κ for labels). The test targets 20–30 listeners; its results are not yet included.

VI. Discussion and Limitations

One mechanism, two uses. Both methods follow from how the model was trained: the prompt was the start of the very utterance the target completes. Ending the prompt at a finished sentence makes "a new sentence" a valid continuation, as at the sentence-boundary cuts of training; a prompt that stops mid-phrase asks the model to finish something it was not given the words for, and short targets come out truncated or

replaced by a replay of the prompt. How often training cuts fell at sentence boundaries we could not measure, because the training manifests are not public. And because the model learned to continue a speaker, it continues that speaker's register and liveliness too — so the prompt can carry delivery the text cannot. The same reasoning applies to any continuation TTS model, and the split of labour — prompt for the body of a sentence, a rule for its boundary tone — to any model trained on text without punctuation.

Limitations. One model family is tuned and evaluated in depth. Perceptual evidence rests on the listening test above; the first check had one listener. Expressiveness is also measured acoustically and through one recogniser whose question rate is itself only 39% on real expressive speech; the expressive reference (ShEMO) is acted radio speech whose texts we attached by recogniser matching (279 of 760 clips) and which has no speaker labels. Style strengths and the extension sentence are hyperparameters chosen on few development renders. A listening test of the whole pipeline is the next step.

VII. Conclusion

A small CPU TTS model can be made both more robust and more expressive without training, through the one input that carries acoustics into the model: its voice prompt. Ending the prompt at the model's own finished sentence halves failed short renders, rescues clips cut mid-utterance and lets the engine drop its retries. Re-styling that sentence makes the model itself speak exclamations and surprise higher and livelier, so the signal processing shrinks to sentence endings and a loudness plan, and questions are heard as questions three times as often at no loss of intelligibility. A two-recogniser consensus and a recogniser used as a listener made both effects measurable. The engine, sets and scripts are available at https://github.com/abolfazltorbat/persian_tts.

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